README: Replication Package

Paper Title

Green or Growth? Unveiling the Eco-environmental Consequence of Cross-border E-commerce Pilots in China

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1. Overview

This replication package provides all the publicly available data, code, and instructions required to reproduce the main empirical results of the above-mentioned research paper. The package is designed to facilitate transparency and enable independent verification of all statistical findings reported in the study.

2. Contents

The replication package includes the following folders and files:

This **replication_data&code folder** contains the replication materials for the empirical analyses presented in the paper. It includes the dataset and Stata code used to reproduce the main results.

Before run the program, please run the 5th line in this do file to set the path that you store the above data. The details are presented as follows:

Table 1:

run line 15-16 in dofile using Data1.dta

Table 2:

run line 18-34 in dofile using Data1.dta

* The growth rate of *EEQ* in 2010-2021, as shown in 0.17% in the Section 4.1 by running line 38-47 in dofile using Data1.dta

* The estimated coefficient of eco on *lnGDP* is 0.013 in the Section 4.1 by running line 49-58 in dofile using Data2.dta

// Data2 additionally includes GDP at city-year level

Table 3 and Figure 3:

run line 61-65 in dofile using Data1.dta

Figure 4:

run line 68-170 in dofile using Data1.dta

Table 4:

run line 173-195 in dofile

Column 1 of Table 4: run line 174-178 in dofile using Data3.dta

// Data3 additionally includes bird diversity at city-year level

Column 2 of Table 4: run line 179-183 in dofile using Data1.dta

Column 3 of Table 4: run line 184-188 in dofile using Data4.dta

// Data4 additionally includes the dummies of confounding policies

Table 5:

run line 197-216 in dofile using Data1.dta

Table 6:

run line 218-236 in dofile using Data1.dta

Table 7:

run line 238-273 in dofile using Data1.dta

Figure 5:

run line 275-291 in dofile using Data1.dta

Figure 6 and Column (1) of Table 8:

run line 293-314 in dofile using Data5.dta

// Data5 only redefines variables name to clearly present Figure 6: Balance Diagnostics of PSM

Column (2)-(6) of Table 8:

run line 316-431 in dofile using Data1 and IVroadacess.dta

Table 9:

run line 431-460 in dofile using Data6.dta, Data7.dta and Data8.dta.

Table 10 and Figure 7:

run line 462-640 in dofile using Data1.dta, Data9.dta, GTFP_SBM.xlsx and GTFP_EBM.xlsx.dta.

Table 11 and Table 12:

run line 642-696 in dofile using Data1.dta and Data10.dta.

3. About the detailed description of datasets and do files with data management in this study

The related datasets and detailed do files with data management is placed into **replication_data management folder.** The dataset ***balanced_panel.dta*** constitutes the original data used in this study.

- The **adjacent_city.dta** is datasets for the adjacent cities.
- The **bird_diversity.dta** is datasets for bird species richness from the China Birdwatching Report, with the original data available at [http://www.birdreport.cn/].
- The **citycode.dta** contains the for code of each city and province.
- The **confounding_policies.dta** include dummy variable for the "*E-Commerce to Villages*", the "*National Ecological Civilization Construction Demonstration Zones*", the "*National Key Ecological Function Zones*" policy, and the "*Shan-Shui Initiative*".
- The **ERI.dta** is datasets for prefecture-year level environmental regulation intensity.
- The **GDP_city.dta** is datasets for prefecture-year level GDP.
- The **log_ind_area.dta** is datasets for land used for industry, logistics and warehousing .
- The **logistics_corporates.dta** is datasets for the registered logistics corporates at prefecture-year level.
- The **migration.dta** is datasets for the various population migration patterns at prefecture-year level.
- The **pollution.dta** is datasets for PM2.5 and PM10.

- The **IVroadacess.dta** is datasets for road kilometer and city area for the original data of IV construction.
- The **SBM_EBM.zip** is a compressed file containing the MATLAB code and raw data for calculating *GTFP_EBM* and *GTFP_SBM*. The specific variable definitions are as follows:
  - ✓ **Input Variables**: (1) Labor input (in1): the natural logarithm of the number of employees at year-end (10,000 persons); (2) Capital input (in2): the natural logarithm of capital stock (10,000 yuan); (3) Energy input (in3): the natural logarithm of electricity consumption (10,000 kWh).
  - ✓ **Desirable Output**: Economic output (out1): the natural logarithm of real GDP (100 million yuan).
  - ✓ **Undesirable Outputs**: (1) Smoke and dust emissions (bad_out1): the natural logarithm of smoke and dust emissions (tons); (2) Wastewater discharge (bad_out2): the natural logarithm of wastewater discharge (tons); (3) Sulfur dioxide emissions (bad_out3): the natural logarithm of SO_2 emissions (10,000 tons).

- **Data1:** The dataset ***balanced_panel.dta*** is executed to impute the missing values and generate Data1.dta.
- **Data2:** GDP of city-year level is merged with Data1 and generate Data2.dta.
- **Data3:** Bird diversity at city-year level is merged with Data1 and generate Data3.dta.
- **Data4:** The dummies of confounding policies are merged with Data1 and generate Data4.dta. The confounding policies include the "*E-Commerce to Villages*", the "*National Ecological Civilization Construction Demonstration Zones*", the "*National Key Ecological Function Zones*" policy, and the "*Shan-Shui Initiative*".
- **Data5**: Data5 only redefines variables name to clearly present Figure 6: Balance Diagnostics of PSM.
- **Data6:** The number of logistics corporates at city-year level is merged with Data1 and generate Data6.dta.
- **Data7:** The logistics and industrial land area at city-year level is merged with Data1 and generate Data7.dta.
- **Data8:** The variable of historical migration patterns at city-year level is merged with Data1 and generate Data8.dta.
- **Data9:** The PM2.5 and PM10 are merged with Data1 and generate Data9.dta. The PM2.5 and PM10 are obtained by the dataset of Wei and Li (2023).

  Wei, J., Li, Z. (2023). ChinaHighPM2.5: High-resolution and High-quality Ground-level PM2.5 Dataset for China (2000-2023). National Tibetan Plateau / Third Pole Environment Data Center. https://doi.org/10.5281/zenodo.3539349.
- **Data10:** The intensity of environmental regulation is merged with Data1 and generate Data10.dta.

4. Notes

The dofile.do file contains detailed comments explaining each step of the empirical procedure, which facilitates replication and understanding.

If any issues arise during replication, please ensure that all dependencies and packages are properly installed. For any questions regarding the code or data, please contact the corresponding author.